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Key Points:

- For the first time, we analyze the city-scale and sub-daily extreme rainfall for the entire country of Japan by using high-quality data sets
- Areal reduction factor values generally decrease with increasing area size and increase with longer duration and are affected by multiple physical phenomena
- Under 1.5 K warming, extreme rainfall will increase more for shorter durations and higher return periods, especially in northern Japan

Supporting Information:

Supporting Information may be found in the online version of this article.

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Spatiotemporal Variability of Current and Future Sub-Daily Rainfall in Japan Using State-Of-The-Art High-Quality Data Sets

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Abstract Using potentially best available rainfall data sets for the entire country of Japan (spatial scales of 1- and 20-km), we analyze the 1–24 hr and city-scale (1–400 km²) extreme rainfalls for both current (2006–2020) and future periods (2081–2109) at 1.5 K global warming scenario, complementing previous work that focuses on either coarse spatial and temporal scales or other warming scenarios (e.g., RCP and 2 K warming scenarios). A peak-over-threshold (POT)-based approach is applied to compute areal reduction factor (ARF) for subsequently establishing intensity-duration-area-frequency (IDAF) curves. Our results reveal that ARF values generally decrease with increasing area size and increase with longer duration and are affected by multiple underlying physical phenomena. Moreover, we find a greater increase in the rainfall intensities for shorter durations and higher return periods, ranging from 9.4% (1-hr) to 6.2% (24-hr), averaged for all return periods and 8.3% (25-year) to 7.3% (2-year), averaged for all rainfall durations. Spatially, extreme rainfall intensities are projected to increase by 8.9% in northern Japan (albeit with a less intense rainfall intensity in the current period), which is greater than the rest of the country (6.8%), underscoring the need to focus more on infrastructure designs in northern Japan. The projected IDAF curves further display an increase in the frequency of extreme events at city-scale, for example, 25-year extreme rainfall events in 2006–2020 would likely be 5–10-year events in 2081–2109. Our results with the state-of-the-art data and implementable approach can be utilized for policymaking to reduce the warming-induced risks in Japan and beyond.

1. Introduction

Extreme rainfall intensities will likely increase in the future due to global warming-induced atmospheric moistening (Trenberth et al., 2003). Particularly, a rapid increase in rainfall intensities is projected for sub-daily scales (1–24 hr), primarily owing to more intense convective events (Zhao, Abhishek, Kinouchi, Ang, & Zhuang, 2022). These increases, combined with multiple anthropogenic factors, such as urbanization and land-use change, would likely lead to more frequent and intense rapidly developing (flash) flood events. This prospect of extreme events that occur with little warning can consequentially cause substantial damage to nature and people, including food reduction, infrastructure damage, economic decline, and more water-related diseases (Fowler et al., 2021; He et al., 2021; IPCC AR6, 2022; Lesk et al., 2020; Xie et al., 2022).

A significant number of studies have recently analyzed the global warming impacts on the sub-daily extreme rainfall intensities (Fadhel et al., 2017, 2021; Hosseinzadehtalaei et al., 2020, 2021; Scoccimarro et al., 2015; Takhellambam et al., 2022a; Zhao et al., 2021), based on the commonly used regional and global climate models under RCP or SSP scenarios (main abbreviations and terminologies are listed in Table 1). However, these models are generally designed to simulate the climate conditions under certain external forcings (e.g., greenhouse gas emissions) rather than a specific degree of warming level at a given time (e.g., 1.5 and 2 K warming at the end of the 21st century under the Paris Agreement). In other words, many studies based on RCP and SSP scenarios (or called transient scenarios, see James et al., 2017) provide limited information about future sub-daily extreme rainfall intensities corresponding to 1.5 and 2 K warming levels (or called stabilized scenarios) at the end of 21st century as proposed by the Paris Agreement. Thus, these studies, by using RCP and SSP scenarios, albeit suitable for mitigation planning on carbon reduction globally, put forward bulk estimates and therefore conceal implications for the regional and local (e.g., a city-scale, 1–400 km²) adaptation strategies (for reducing risks from sub-daily extreme rainfall) as required by the two Paris Agreement goals. Furthermore, the limited number of

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climate models used in many studies dealing with regional and local scales (Hosseinzadehtalaei et al., 2020; Laux et al., 2021; Ngai et al., 2020; Supari et al., 2020; Tangang et al., 2019; among others), provide somewhat low confidence in their results, which is mainly because that limited models cannot separate the natural variability (i.e., noise) of the climate system and climate change signals informed by global warming (Aalbers et al., 2018). Natural variability plays a particularly important role in estimating extreme rainfall intensities (Milinski et al., 2020; Suarez-Gutierrez et al., 2021). To reduce this impact from natural variability, Ishii and Mori (2020) demonstrated that at least 30 climate model simulations are needed. Therefore, several research projects have been conducted to provide researchers with a large ensemble of climate model simulations, such as ClimEx (Leduc et al., 2019), Max Planck Institute Grand Ensemble (MPI-GE) (Maher et al., 2019), and a database for policy decision making for future climate change (d4PDF, available from <https://diasjp.net/en/>) (Mizuta et al., 2017), all of which consists of at least 50 climate model simulations. These large ensemble climate model data sets can help constrain the uncertainty arising mainly from internal variability and imperfect model performance in extreme rainfall intensities around the world (Endo et al., 2017; Fujita et al., 2019; Innocenti et al., 2019; Martel et al., 2020; Raymond et al., 2022, among others). However, most of these studies focus on either the extreme rainfall corresponding to relatively coarse temporal (e.g., daily) and spatial (e.g., 10,000 km²) scales or the warming scenarios not targeting two Paris Agreement goals. A detailed analysis of the future sub-daily (i.e., 1–24 hr) extreme rainfall, for a city-scale (e.g., 1–400 km²) and a specific warming level (e.g., 1.5 K) at the end of the 21st century, by using a large ensemble of climate models (e.g., >30) is still limited.

Regarding the observed sub-daily rainfall data sets, a handful of studies have used the observations provided by the rain gauges, satellites, and radars (Hosseinzadehtalaei et al., 2020, 2021; Marra & Morin, 2015; Minh et al., 2022; Takhellambam et al., 2022b; Zhao, Abhishek, & Kinouchi, 2022), and the reanalysis data sets (e.g., ERA5) (e.g., Hosseinzadehtalaei et al., 2020; Zhao, Abhishek, Kinouchi, Ang, & Zhuang, 2022). However, these data sets have inherent limitations, such as limited spatial representativeness of rain-gauge data sets, poor representation of convective process (e.g., reanalysis data sets), and quantitative precipitation estimation (QPE) uncertainty in radars. To solve these problems, precipitation products were improved by merging radar data with rain-gauge data (Ochoa-Rodriguez et al., 2019). For example, the Radar/Raingauge-Analyzed precipitation data sets at a time interval of 30 min and spatial interval of 1-km have proven to be a useful and efficient tool for understanding the characteristics of sub-daily extreme rainfall on the regional and local scale (Ishizaki & Matsuyama, 2018; Kato, 2020; Ochoa-Rodriguez et al., 2019; Z. Zhou et al., 2019). In densely populated cities like Tokyo and Osaka in Japan, where these cities are fully urbanized with complex and somewhat deteriorated underground spaces (e.g., subway stations) and located in a low-lying terrain (Nakasaka & Ishigaki, 2021), a holistic quantification of the sub-daily extreme rainfall and the associated risks by using high-quality rainfall data sets, such as those by merging radar and rain-gauge observations, becomes crucial.

The sub-daily extreme rainfall data is commonly used to generate intensity-duration-area-frequency (IDAF) curves on a city scale for designing hydraulic structures. IDAF curves are the extension of the conventional intensity-duration-frequency (IDF) curves and estimate the average rainfall intensity in a given rainfall duration and return period considering the affected area size (Mélèse et al., 2019; Zhao, Abhishek, Kinouchi, Ang, & Zhuang, 2022). The IDAF curve includes an important ingredient—the areal reduction factor (ARF), which refers to the ratio between areal rainfall intensity and point rainfall intensity. In the literature, depending on how rainfall data sets and target areas are selected, two ways are prevalent to compute ARF, namely, “fixed-area” and “storm-centered” approaches (Biondi et al., 2021). Both approaches have their advantages and disadvantages (Biondi et al., 2021; Olivera et al., 2008). On the one hand, the “storm-centered” approach, estimated for an individual storm event, is only applicable to specific storm events and does not account for the frequency of heavy rainfall (Veneziano & Langousis, 2005). On the other hand, the “fixed-area” approach, relying on the return period of heavy rainfall over a fixed area, is perhaps the most widely used in hydraulic infrastructure designs (e.g., US weather Bureau method, United Kingdom Flood studies report, Bell's method, among others) (Biondi et al., 2021; Pavlovic et al., 2016).

Considering the return period, most studies on ARF have adopted the block maximum (BM) approach (Blanchet & Mélèse, 2020; Breinl et al., 2020; Mélèse et al., 2019; Panthou et al., 2014; Pavlovic et al., 2016, among others), which is because that BM approach can ensure the independence of the maximum series and is easy to implement. However, the BM approach only samples one maximum rainfall value per block (e.g., per year), which undermines inferences for the regions with insufficient data record length. This simplicity in sampling could result in an inadequate representation of the rainfall events and loss of information in the following cases,

Table 1
Mainly Used Abbreviations and Terminologies

Abbreviations	Full name/terminology
ARF	Areal reduction factor
Areal rainfall	Rainfall information covering a certain area
BC	Bias correction
BM	Block maximum
Consistent pattern assumption	Spatial rainfall patterns remain consistent from current to the future period
d4PDF	database for policy decision-making for future climate change
Return rainfall	Rainfall amount with a given return period and duration
GPD	Generalized Pareto Distribution
ID(A)F	Intensity-duration-(area)-frequency
obs	Observation
Point rainfall	Rainfall information at one point (rain-gauges)
POT	Peak-over-threshold
RA	Radar/Raingauge Analyzed Precipitation
RCP	Representative Concentration Pathway
SSP	Shared Socioeconomic Pathways
TCs	Tropical cyclones

(a) when non-sampled rainfall values in some years may be larger than the sampled annual maxima of other years, (b) when the study period includes dry years with very low yet annual maximum values (Tabari, 2021). To resolve this problem, the peak-over-threshold (POT) approach, which additionally focuses on the rainfall values larger than a given threshold, has been applied by many researchers in a multitude of fields, such as frequency analysis of rainfall and flood (Hosseinzadehtalaei et al., 2020; Tabari, 2021), environmental and economic issues (Coles et al., 2001). The POT approach outperforms the traditional BM approach in frequency analysis, particularly in small sample sizes, because it can use more extreme information. However, the POT approach has been limitedly explored in the development of ARF and is more restricted to local areas (e.g., Le et al., 2018; Tabari, 2021; Vrbán et al., 2018). Therefore, it merits consideration to investigate the capability of POT to derive ARF values and the applicability of ARF to future spatial downscaling on large scales, such as national scales, which will subsequently facilitate the policymaking for adaptation strategies in a changing climate.

This study has considered both the rainfall data issue (i.e., specific warming level, spatial and temporal scales, accuracy) and the method issue (i.e., POT approach). As a whole, the purpose of this study is to establish the current and future (at a specific warming level, see Section 2.2) city-scale IDAF curves based on high-resolution data sets (total data volume amounting to approximately three terabytes, as described in Section 2.1), and a POT approach (Section 3). To the best of our knowledge, this is the first study to analyze the city-scale and sub-daily extreme rainfall for the entire country of Japan, and such extreme rainfall is the first time to be investigated under 1.5 K warming scenarios. Our findings and methodologies can foster policy interventions for water resource management to reduce the warming-induced risks in Japan and beyond.

The remainder of this paper is organized as follows. Section 2 introduces the study area and rainfall data sets. A framework for establishing IDAF curves (i.e., IDF curves with different spatial scales) by using the POT approach is described in Section 3. The warming-induced changes in sub-daily extreme rainfall and IDAF curves are presented in Section 4. At last, Section 5 summarizes the findings.

2. Study Area and Data Sets

2.1. Study Area

Japan, located in eastern Asia, is the fourth largest island country (geographical area~378,000 km²), bounded to the north by the Sea of Okhotsk, to the west by the Sea of Japan, to the southwest by the East China Sea as

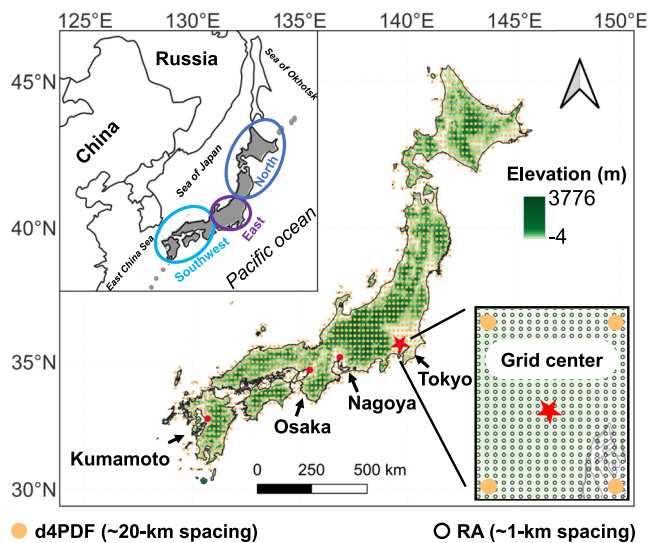


Figure 1. The location and elevation (above mean sea level) of the study area and grid points of d4PDF (1,308 grid points) and RA data (>500,000 grid points). RA for the whole of Japan (the Nansei islands, the Northern Territories and some remote islands were not considered in this study) is not shown, and the sizes of shapes (i.e., circles, stars) were adjusted for better visualization. Northern (from Fukushima to Hokkaido prefecture in terms of latitude), eastern (from Aichi to Niigata prefecture), and southwestern (the remaining regions) Japan are displayed in the upper-left figure. The detailed data information is listed in Table 2.

well as to the east and south by the Pacific Ocean (Figure 1). Its terrain is mostly rugged, and over 80% of the land surface is covered by mountains, with the highest elevation of 3,776 m in Mount Fuji. Japan's climate is mainly influenced by the seasonal monsoon circulation, where eastward winds pick up moisture from the Sea of Japan and deposit the moisture mainly in the format of snow on the Sea of Japan side in winter (December–February); on the other hand, during the summer (June–August), air movements from the Pacific Ocean bring high moisture to most of the country, leading to frequent heavy rainfall events during this season. Apart from the monsoon's effects, high moisture brought by the long-existing “Baiu front” (i.e., frontal rainfall) in the early summer, followed by the tropical cyclones in the mid-summer to early autumn, results in abundant and heavy rainfall events (Hirokawa, Kato, Araki, & Mashiko, 2020). For better discussion, four case study cities, Tokyo, Nagoya, Osaka, and Kumamoto, were selected. Tokyo and Osaka are regarded as the largest and tenth cities by population in the world, with approximately 37 and 19 million inhabitants, respectively (United Nations, 2018). Nagoya ranks ~4th in terms of population in Japan (Suzuki & Nijkamp, 2018). These three cities play a fundamental role in the economic activities and manufacturing industry in Japan (Suzuki & Nijkamp, 2018). Unlike these cities, Kumamoto is considered here because it is located on the most south-westerly island that witnesses frequent typhoons and subsequent landfalls (Aoki, 1985). Such high population, economic contribution, and high risks necessitate a better understanding of global warming impacts to avoid potential warming-induced risks (e.g., urban flooding).

2.2. Rainfall Data Sets

The potentially most accurate (in terms of data sources, temporal and spatial scales) precipitation data set in Japan, with the full name of “Radar/Raingauge Analyzed Precipitation” (Hereinafter, RA), was used as reference or observed data. The spatial and temporal scales of RA are 1-km $\times$ 1-km (0.0125° $\times$ 0.00833°) and 1-hr (reconstructed from 30-min data sets), respectively, with over 500,000 grid cells covering the entire country of Japan (excluding the Nansei islands, the Northern Territories and some remote islands) from 2006 to 2020. RA was developed by combining rainfall information from 10,000 rain gauges and radar observations based on rigorous quality control and calibration (Nagata, 2011). The procedure of producing RA mainly includes five steps (Nagata, 2011), (a) radar intensity data are obtained by accumulating the observed echo intensity data; (b) a quality-controlled approach is followed to ensure that the average of accumulated data over a given area (and grid cell) equals to that of all other radars (and rain gauges) covering the same area (and grid cell); (c) the accumulated radar data are calibrated by adjusting their distribution to resemble the distribution of rain gauge data; (d) the calibrated data are further composited for the entire country of Japan; (e) The values for the grid cells where rain gauges are located are replaced with the rain gauge precipitation values. Therefore, the merged RA data, developed by leveraging the rain gauges and radar data (Ochoa-Rodriguez et al., 2019), usually perform better than rain gauges (in terms of spatial coverage) and remotely sensed data sets (in terms of accuracy) alone, as well as the commonly used reanalysis data sets (e.g., JRA-55, ERA5, CFSR, among others) (Ang et al., 2022; Fukui et al., 2018). Owing to the computational reason, for the results of the whole of Japan, we restricted the number of RA grid cells by selecting the nearest 25 RA grid cells to the d4PDF grid points (shown in Figure 1), ending up with 32,700 (=1,308 $\times$ 25) RA grid cells.

Regarding the future rainfall data sets, we used regional climate simulations provided by the “database for policy decision-making for future climate change” (hereinafter, d4PDF). The d4PDF data sets are simulated by using a 60-km global atmospheric model (AGCM) and a 20-km regional climate model (NHRCM) covering the Japan area (Mizuta et al., 2017). Three different sets of experiments are performed: historical climate simulations, historical non-warming simulations, and future climate simulations, all of which have 54 to 100 ensemble members based on different initial conditions, lower boundary conditions, and CMIP5 models. Historical climate simulations are performed by using the observed sea surface temperature (SST), sea-ice concentration (SIC), and sea-ice thickness (SIT) from COBE-SST and observed global-mean green gas concentrations. The historical

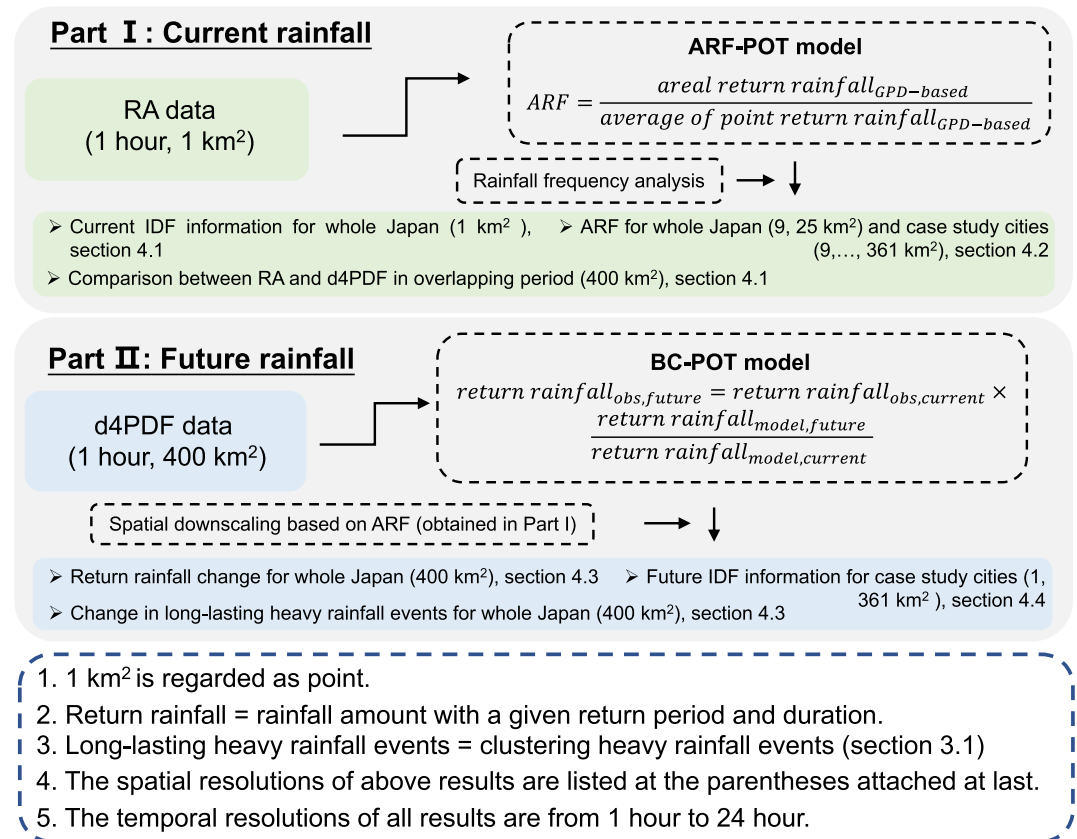


Figure 2. A flowchart of the research framework. For simplicity, return rainfall change at 400 km² was regarded as the change at 361 km² to match with the areal reduction factor in four case study cities.

non-warming simulations are performed by using the same boundary conditions as historical climate simulations, except that the observed warming trend is removed. Future climate simulations consist of three fixed global warming levels, +1.5 K, +2 K, and +4 K, at the end of the 21st century (relative to preindustrial levels). To ensure such fixed warming levels, the climatological SST warming patterns are scaled by multipliers and added to the observed SST data after removing the long-term trend (Ishii & Mori, 2020). In the present study, we used the d4PDF outputs, consisting of 50 ensemble members for the historical reproduction (1981–2010) and 54 members for the future 1.5 K warming future projection (2081–2109), with a temporal scale of 1-hr and spatial scale of 20-km (Figure 1). The choice of the 1.5 K warming scenario (accessible to the public from February 2022, https://www.miroc-gcm.jp/d4PDF/index_en.html), is ascribed to the relatively fewer studies on 1.5 K than studies on 2 and 4 K and growing global awareness on 1.5 K warming (Duan et al., 2021), thus providing some new aspects on future extreme rainfall and associated risks. Note that, “extreme” here denotes the rainfall amount that deviates from the normal rainfall amount. For computational feasibility and reducing the snow impacts, we only extracted the rainfall data sets for rainy seasons (April–October) from RA data and d4PDF data.

3. Methods

The research framework described in Figure 2 introduces the approaches to obtain current and future sub-daily return rainfall information mainly through rainfall frequency analysis and spatial downscaling.

1. Current IDF information, with spatial scales of 1 km², was obtained based on RA data for the entire country of Japan (Sections 3.1 and 4.1).
2. Using RA data, we computed the ARF values for the entire country of Japan and four case study cities (Sections 3.2 and 4.2).
3. ARF values obtained in Step 2 were used to spatially downscale future return rainfall based on d4PDF (Section 3.3).

Table 2

List of Precipitation Data Sets Used in This Study

	RA	d4PDF (1.5 K warming)
Temporal scale	1-hr (aggregation from 30-min)	1-hr
Spatial scale	1-km (1 km ²)	20-km (400 km ²)
Spatial coverage	Whole Japan	Whole Japan
Developer/Supporter	Japan Meteorological Agency	Ministry of Education, Culture, Sports, Science, and Technology
Ensemble members	–	50 for historical 54 for future
Time period	Current (2006–2020)	Historical (1981–2010) Future (2081–2109)
Reference	Hirokawa, Kato, Tsuguti, and Seino (2020) and Nagata (2011)	Mizuta et al. (2017) and Ishii and Mori (2020)

- Percentage changes in return rainfall relative to the historical period were obtained for the entire country of Japan (Section 4.3).
- For case study cities, we established the IDF curves for spatial scales of 1 and 361 km² by using the IDF information in Step 1, ARF values in Step 2, and changes of return rainfall in Step 4 (Section 4.4).

3.1. Rainfall Frequency Analysis

The cumulative distribution of GPD for fitting POT is given below (Coles et al., 2001),

$$F(x) = \begin{cases} 1 - \left(1 + \frac{\xi(x-u)}{\sigma}\right)^{-\frac{1}{\xi}} & \text{for } \xi \neq 0 \\ 1 - \exp\left(-\frac{x-u}{\sigma}\right) & \text{for } \xi = 0 \end{cases} \quad (1)$$

where x is the POT value, σ and ξ denote the scale and shape parameter. u denotes the threshold value, over which rainfall amount is regarded as the POT value. A high enough value, the 99.8th percentile of all rainfall time series, was chosen as the threshold to ensure the asymptotic assumption of GPD. GPD assumes that extreme rainfall events should happen randomly following the Poisson process. In other words, extreme events are independent, a condition that is not always fulfilled when dealing with the POT value (Coles et al., 2001). For example, extreme daily rainfall events may last for a few days. To avoid the dependence of POT values (extreme events tend to appear in clusters), a time interval of 24 hr was applied to select the POT values, following the suggestions from Hosseinzadehtalei et al. (2020) and Willems (2009). Moreover, the cluster maxima were used to further ensure the independence assumption suggested by Coles et al. (2001). The estimation of these parameters was implemented using the “POT” package available in R software (<https://cran.r-project.org/web/packages/POT/POT.pdf>).

The extremal index of GPD is used to measure the degree of clustering of the extreme rainfall events (Coles et al., 2001). The reciprocal of the extremal index can be seen as the average size of the clusters. For example, extreme rainfall events with a “extremal index = 0.5 (or 0.2)” denote, on average, two (or five) extreme rainfall events happen continuously. The estimation of the extremal index was implemented through the *exi* function from the “EVD” package provided by R software (<https://cran.r-project.org/web/packages/evd/evd.pdf>).

3.2. ARF-POT Model

The computation of ARF is modified from Bell's method (Le et al., 2018), and mathematically represented as,

$$\text{ARF} = \frac{\text{areal return rainfall}}{\text{average of point return rainfall}} \quad (2)$$

where, areal return rainfall denotes the return rainfall based on areal rainfall amount (spatial aggregation from point rainfall amount), and average of point return rainfall is calculated based on the point rainfall amount observed in the target area. ARF based on Equation 2 could be used as a proxy for the spatial variability of the

rainfall field in the area of interest (where a larger spatial variability mostly denotes a smaller ARF). However, we need to be cautious that spatial aggregation can cause some systematic bias in estimating this spatial variability (see Section 4.2). The arithmetic mean method was used for spatial aggregation while taking into account the evenly distributed RA grid points. Due to computational constraints, we only considered squared area sizes of 9 and 25 km² for the entire country of Japan. The squared area sizes of 9, 25, 49, 81, 121, 169, 255, 289, and 361 km² were further investigated in four case cities (i.e., Tokyo, Nagoya, Osaka, and Kumamoto). Although other area shapes (e.g., circular) can also be used for spatial aggregation, they have limited impacts on the ARF (Kim et al., 2019; Mélése et al., 2019). Please note that the ARF values based on squared area size may partly deviate from the true values, as this squared-area approach does not fully consider the physical characteristics of the target area and the rainfall.

The return rainfall is estimated based on the “N-year return level” of GPD (Coles et al., 2001) which is given as,

$$\text{return rainfall}_N = \begin{cases} u + \frac{\sigma}{\xi} [(N n_y \zeta_u)^\xi - 1] & \text{for } \xi \neq 0 \\ u + \sigma \log(N n_y \zeta_u) & \text{for } \xi = 0 \end{cases} \quad (3)$$

where, N is the considered return period (i.e., 2-, 5-, 10-, 25-year), return rainfall_N is the return rainfall that is expected to happen once every N years, n_y is the number of observations in each year, and ζ_u is the ratio of observations exceeding u (i.e., cluster maxima). The return rainfall of five durations (i.e., 1, 3, 6, 12, and 24 hr) was obtained to compute the ARF.

3.3. BC-POT Model

The d4PDF data sets need to be bias-corrected due to the limited representation by the model (Laux et al., 2021). However, the commonly used BC methods of quantile mapping have been criticized for altering the projected climate change signals and spatiotemporal consistency of target variables due to some weak assumptions (Ehret et al., 2012). Although multiple improved BC methods have been proposed (e.g., Chen et al., 2018; Grillakis et al., 2017; He et al., 2016; L. Zhou et al., 2022), these methods have limitations mainly due to a lack of robust physical validation, which impairs their generalization to other target variables or regions (Ghimire et al., 2019; Hnilica et al., 2017; Kay et al., 2021). Therefore, we introduce a simple BC method based on the idea of “delta change” or “change factor” (Zhao et al., 2021) in a POT framework,

$$\text{return rainfall}_{\text{obs, future}} = \text{return rainfall}_{\text{obs, current}} \times \frac{\text{return rainfall}_{\text{model, future}}}{\text{return rainfall}_{\text{model, current}}} \quad (4)$$

so that the replacement of return rainfall in Equation 3 with Equation 4 can help obtain the observed return rainfall for the future period. In other words, by using the relative change informed by the climate models instead of the absolute values, this approach can avoid potential issues caused by the bias in these absolute values. Notably, return rainfall change is computed using d4PDF for both future and historical periods, as it is not available for the current period (Table 2). Nevertheless, for the selected four cities, this return rainfall change was used directly (based on Equation (4)) to obtain future return rainfall because a preliminary analysis based on additional observed data sets (not shown) found that return rainfall in the four cities almost remains unchanged from historical period to the current period.

As the spatial resolution of d4PDF is 400 km², the $\text{return rainfall}_{\text{obs, future}}$ further needs to be spatially downscaled to smaller spatial scales (e.g., 1 km²) for practical applications of different hydraulic infrastructure designs. Although some researchers have used the d4PDF with the spatial resolution of 25 km² (NILIM, 2022), the restricted data access and the lack of a 1.5 K warming scenario motivate us to apply statistical spatial downscaling methods. In this article, the spatial downscaling was implemented through the reciprocal of ARF, where the return rainfall at fine spatial scales (e.g., 1 km²) equals the return rainfall at coarse scales (e.g., 400 km²) divided by the ARF (shown in Equation 2). In other words, $\text{Return rainfall}_{\text{fine spatial scales}} = \text{Return rainfall}_{\text{coarse spatial scales}} \div \text{ARF}$. Note that although the spatial variability of rainfall events may compromise the reciprocal use of ARF for spatial downscaling, this approach can be acceptable since this downscaling method can estimate mean extreme rainfall reasonably well (Peleg et al., 2018), and these cities are located on relatively flat terrain. In addition, the correction factor was used to remove the bias of fixed-interval measurement (Zhao, Abhishek, & Kinouchi, 2022), an approach used to convert RA and d4PDF data from clock-hour values to true values.

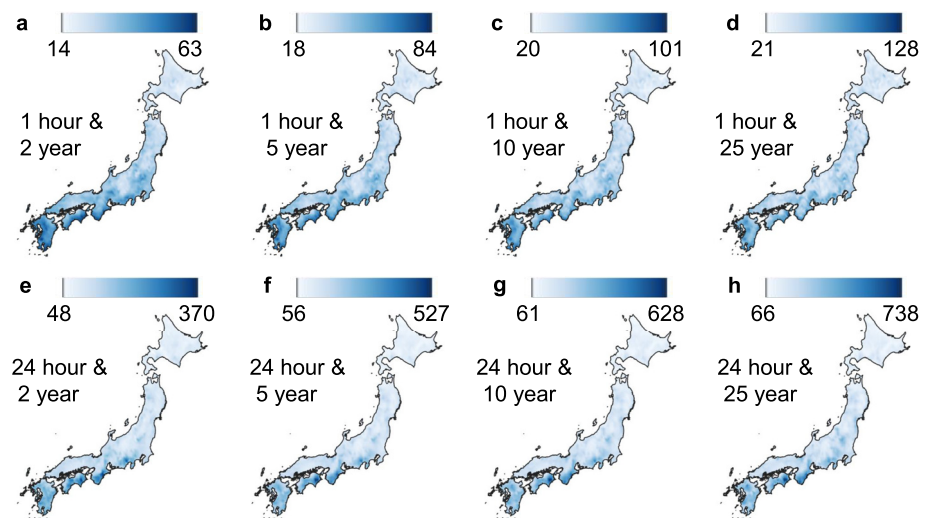


Figure 3. Plots showing the (a–d) 1- and (e–h) 24-hr rainfall amount (mm) (spatial scale of 1 km²) for the return period of 2-, 5-, 10-, and 25-year during the current period (2006–2020). The rainfall amount in each panel was obtained based on Equation 3. The color tables have been adjusted for each plot for better visualization. The plots for 3-, 6-, and 12-hr are shown in Figure S1 of Supporting Information S1.

4. Results and Discussion

4.1. RA Based Current Return Rainfall and Comparison With d4PDF

Figure 3 and Figure S1 in Supporting Information S1 display the return rainfall based on RA data with the spatial scales of 1 km² in Japan. We observe that return rainfall generally decreases from southwest to north for all return periods varying from 2-year to 25-year and rainfall durations of 1–24 hr. Relatively heavy rainfall is observed on the Pacific side of both southwestern and eastern Japan, especially for the longer rainfall durations, whereas a small amount of rainfall is seen on the Japan sea side. The underlying phenomena are possible that (a) the tropical cyclones (TCs), which contribute approximately 40% to heavy rainfall events in Japan, generally become weaker when they move northward (Mori et al., 2021), and (b) the mesoscale convective systems (including the “Baiu front”) usually bring heavy rainfall in southwestern Japan and Pacific sides (Hirockawa, Kato, Tsuguti, & Seino, 2020). Owing to these phenomena, heavy rainfall events in southwestern and eastern Japan (especially for the shorter rainfall durations and Pacific sides) are more prone to happen continuously than the events in Northern Japan (Figure S2 in Supporting Information S1). Moreover, the coefficient of variation (CV) for each plot in Figure 3 and Figure S1 in Supporting Information S1 is found to increase with the rainfall durations and return periods, ranging from 29% (1-hr and 2-year) to 44% (24-hr and 25-year). This high fluctuation in heavy rainfall events with a low frequency and long duration can be ascribed to the extremely large values in southwestern and eastern Japan, underscoring the significance of TCs and mesoscale convective systems in Japan.

The comparison between RA and d4PDF in Figure 4 indicates that d4PDF can reproduce well the spatial distribution of heavy rainfall in Japan, where large rainfall intensities are in southwestern and eastern Japan, especially on the Pacific side. However, bias exists in the values of rainfall amount and varies among different rainfall durations and regions. Temporally, d4PDF tends to underestimate the heavy rainfall with short rainfall durations with an average value of −33.6%, 3.9%, and 9.4% for the duration of 1-, 6-, and 24-hr, respectively, which can be explained by the convective parameterization (playing major roles in short duration heavy rainfall) (Kawase et al., 2019). Spatially, a larger bias (bias = 32.2%, absolute values averaged over the rainfall durations of 1-, 6-, and 24-hr) is observed in southwestern and eastern Japan against the northern region of 27.2%. As mentioned above, heavy rainfalls in southwestern and eastern Japan are dominated by the TCs and mesoscale convective systems, both of which are insufficiently resolved in the d4PDF due to computational constraints. Specifically, the sea–air coupling process in d4PDF can be affected by the fixed SST (Section 2.2), which may further affect the TC characteristics (Ogata et al., 2015), subsequently leading to the relatively large bias in heavy rainfall in southwestern and eastern Japan. Therefore, multi-model high-resolution simulations with the ability to fully resolve the TCs and convective systems are pivotal to reducing both the error and uncertainty in heavy rainfall

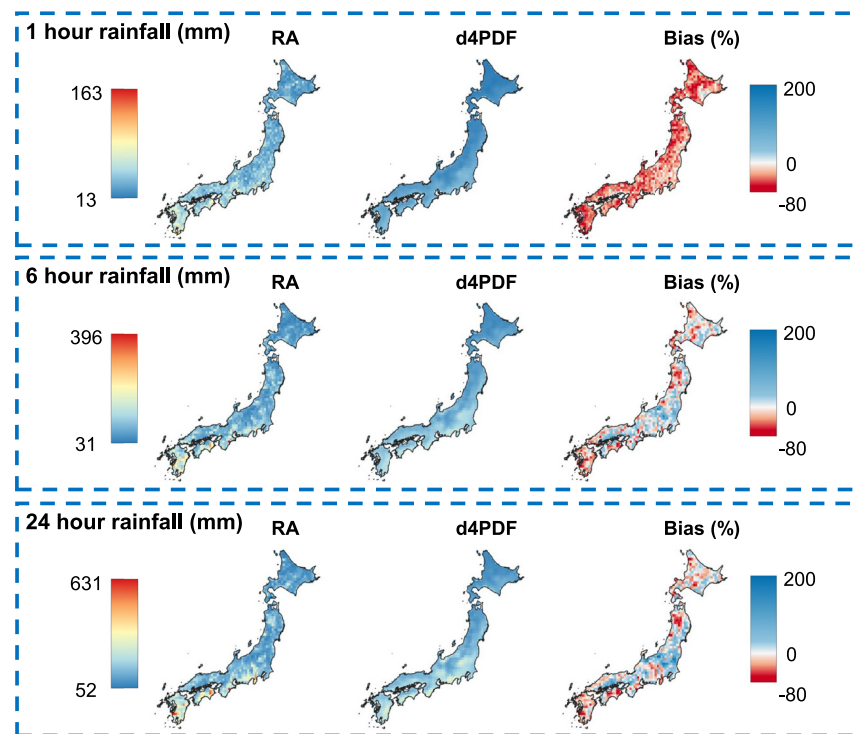


Figure 4. The spatial pattern of (top) 1-, (middle) 6-, and (bottom) 24-hr maximum rainfall for (left) RA data, (center) d4PDF, and (right) associated bias $((d4PDF - RA)/(RA))$ during the overlapping period (2006–2010). For consistent comparison, the RA grid cells were spatially aggregated to match with d4PDF grid cells.

projection. In all, our findings highlight the need for synergistic use of both observed and projected data sets, which can consider the TCs and convective systems, to assist policymakers in implementing sustainable adaptation strategies in Japan and other east Asia regions.

4.2. ARF in Space and Time

To obtain ARF values for the squared size of 9 and 25 km² in the entire country of Japan, the ARF-POT model (Equation 2) was applied by using RA data, as shown in Figure 5 and Figure S3 in Supporting Information S1. An inverse relationship between ARF and elevation is evident for the elevation below 1,000 m, whilst the relationship is extremely weak for the elevation above 1,000 m. The physical mechanisms underlying this difference are very complex. On the one hand, the region with a higher elevation is usually characterized by a larger spatial variability of rainfall field (e.g., orographic rainfall) (Buytaert et al., 2006). On the other hand, as elevation further increases, the orographic rainfall may occur less owing to limited moisture. Instead, some large-scale and less spatially varying storms (such as frontal storms) may leave traces inside the squared area, reducing the spatial variability of the rainfall field, eventually increasing the ARF values (Kim et al., 2019). However, note that ARF values are also affected by the physical characteristics of the watershed in Japan. For example, in this island country, the coastal areas can be seriously affected by sea-land interactions (Hirockawa, Kato, Araki, & Mashiko, 2020). Moreover, in southwestern Japan, high ARF values are more observed than in other regions (Figure 5 and Figure S3 in Supporting Information S1), which is mainly owing to the frequent mesoscale convective systems and large-scale TCs in the region (decreasing the spatial variability of rainfall field and increasing the ARF) (Hirockawa, Kato, Tsuguti, & Seino, 2020; Mori et al., 2021).

In the case study cities, for a fixed rainfall duration (e.g., 1-hr), as shown in Figure 6a, ARF values are found to decrease with the area size. This declining trend is because the heterogeneity level of rainfall field increases as the area size exceeds the storm trace scales, subsequently reducing the ARF values. This declining trend, however, becomes less evident as the duration increases (e.g., 1-hr and 24-hr in Figure 6c), which could be explained by the phenomenon that rainfall events with longer durations (e.g., mesoscale rainfall systems) are mostly less

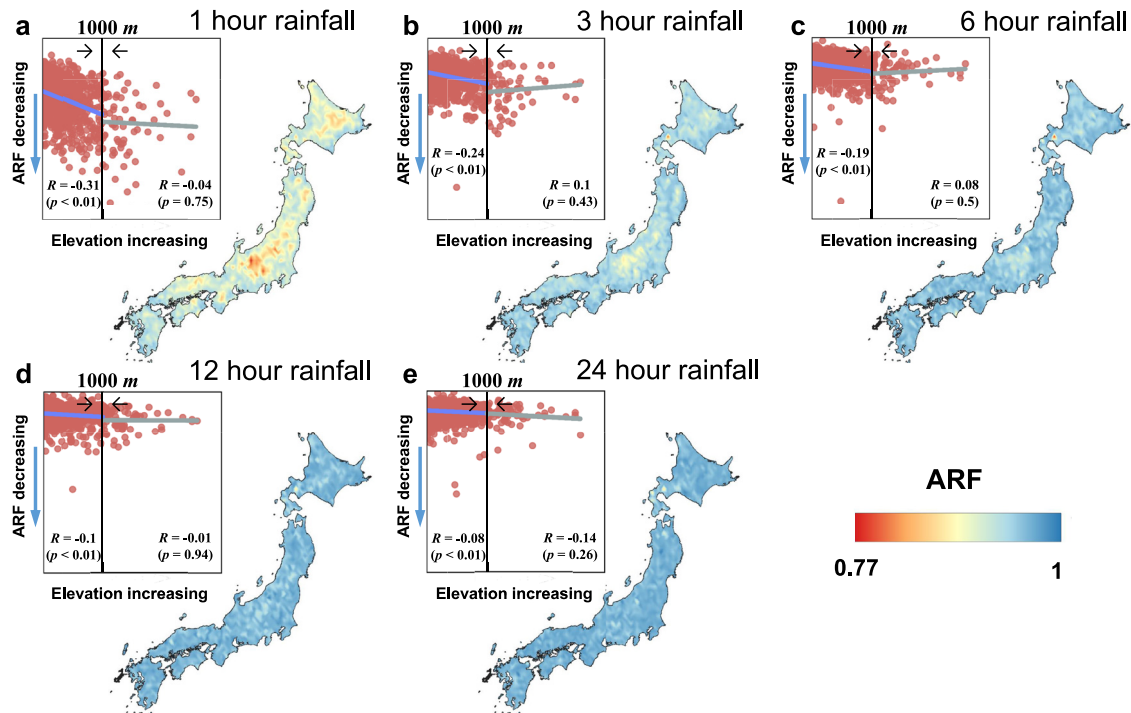


Figure 5. Spatial distribution of areal reduction factor (ARF, 25 km²) based on RA data for the rainfall duration of (a) 1-, (b) 3-, (c) 6-, (d) 12-, and (e) 24-hr. ARF values were calculated after taking the median values of all return periods. The scatter plots give the relationships between ARF and elevation (above mean sea level). The plots of ARF (9 km²) are shown in Figure S3 of Supporting Information S1.

spatially varying than the events with shorter durations (e.g., localized rainfall systems). This phenomenon also explains why ARF values for a fixed area size (Figures 6a and 6c) generally increase with rainfall durations. However, there are some exceptions, for example, ARF decreases from 0.99 in 6-hr to 0.98 in 12-hr for the case of 9 km² in Tokyo (Figure S4 in Supporting Information S1). These exceptions possibly arise from the confounding impacts of complex urban terrain and individual or coexisting rainfall patterns (e.g., convective and frontal rainfall) (Hirockawa, Kato, Araki, & Mashiko, 2020; Nakasaka & Ishigaki, 2021). The relationships between ARF and return periods have not yet been quantified in the literature; both inverse relationships and negligible relationships were reported (e.g., Breinl et al., 2020; Le et al., 2018; Wright et al., 2014). Our results suggest that ARF varies with the return period for the rainfall durations of 1- and 3-hr (Figure 6f). This phenomenon is possible because of the impacts of convective activities on the extreme rainfall events within short durations (e.g., more convective activities = lower ARF). However, no significant relationships exist for the longer durations (Figure 6f). This is mainly because the long-lasting extreme rainfall events can be triggered by complex rainfall generation systems (e.g., local scale and mesoscale systems), which interact differently with the orography (Ma et al., 2022; Marra et al., 2021), complicating the relationship between ARF and return period. In a nutshell, our results found that ARF generally decreases with the area size and increases with the duration, and elevation has a limited impact (<10%) on the variance in ARF values. For a better understanding of the spatial variability of ARF values, multiple other underlying physical phenomena (e.g., watershed location) need further quantitative assessment in future work.

4.3. Change in Return Rainfall and Extremal Index

Spatially, the relative change (future-historical/historical) in the return rainfall (Figure 7 and Figure S6 in Supporting Information S1) is found to be greater (8.9%) in northern Japan than in other regions (6.8%), when averaged over all rainfall durations and return periods. This spatial finding is generally consistent with the spatial patterns (denoting more increase in heavy rainfall in Northern Japan than other Japanese regions) obtained by Osakada and Nakakita (2018), Nosaka et al. (2020), and Watanabe and Utsumi (2018), although all of these works were based on other warming scenarios (e.g., 4 K) and relatively coarse temporal scales (e.g., daily rainfall). This

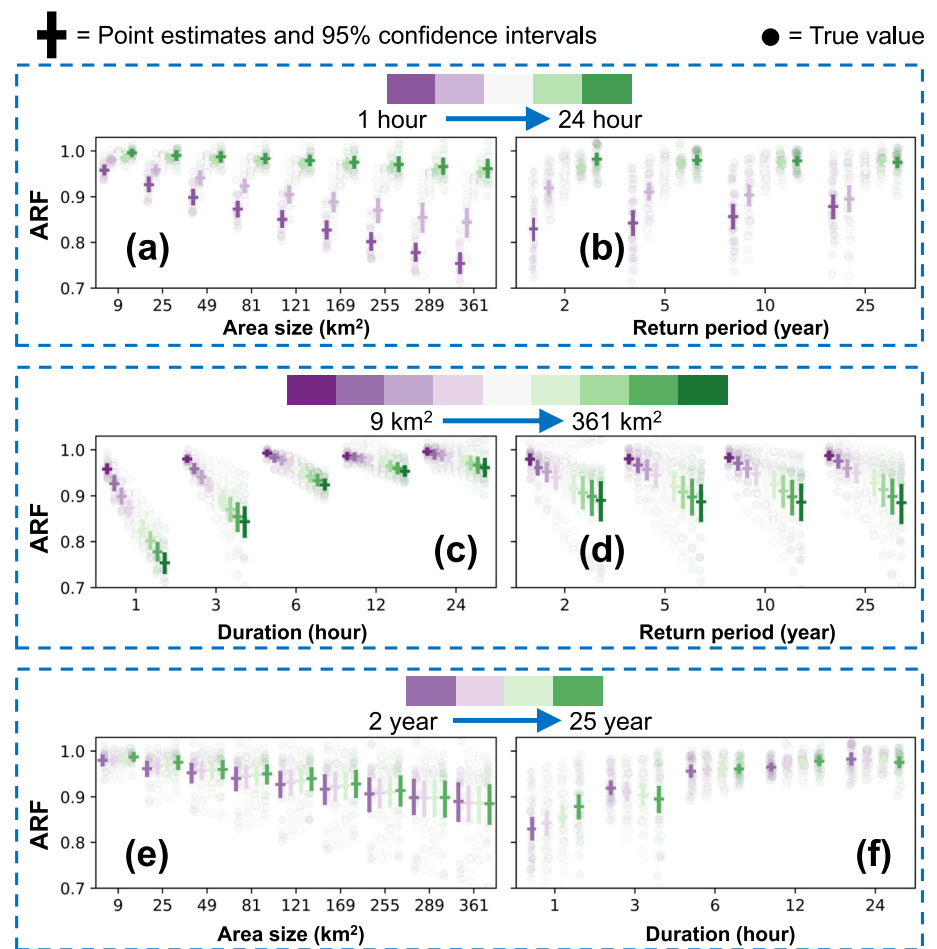


Figure 6. Plots showing the areal reduction factor (ARF) comparison between (a and b) different rainfall durations, (c and d) area sizes, and return periods (e and f) for the case study cities. The ARF values were estimated based on Equation 2. In a, to remove the impacts of the return period, ARF values were calculated after taking the median values of all return periods. A similar procedure was also adapted to other subplots for a better comparison. Point estimates are the ensemble median of all ARF values. The plots for each city are shown in Figures S4 and S5 of Supporting Information S1.

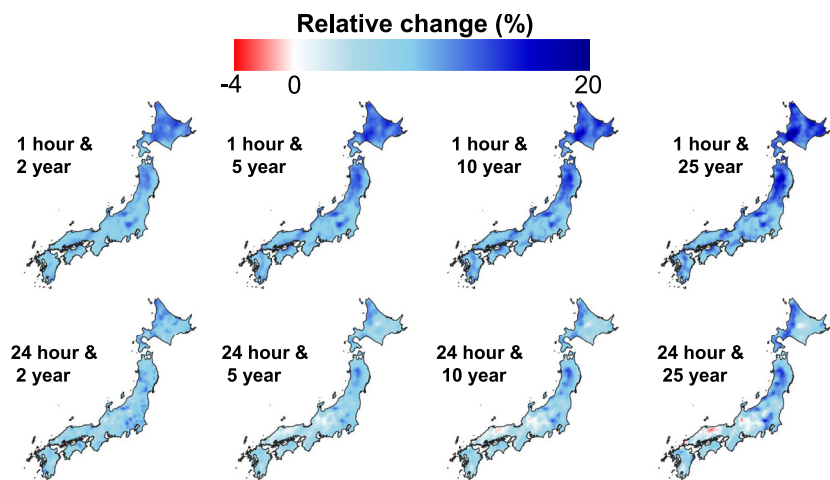


Figure 7. The projected changes in (top) 1- and (bottom) 24-hr heavy rainfall based on the ensemble mean of d4PDF for the future period (2081–2109) relative to the historical period (1981–2010). The changes in 3-, 6-, and 12-hr heavy rainfall are shown in Figure S6 of Supporting Information S1.

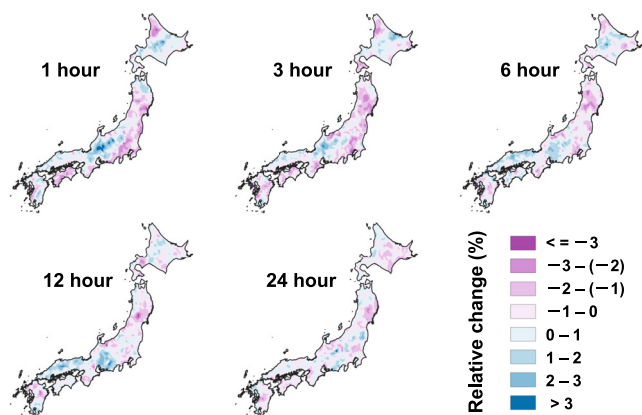


Figure 8. Same as Figure 7 but for long-lasting (clustering) heavy rainfall events. The estimation of clustering heavy rainfall events was based on the extremal index of GPD.

spatial difference is mainly due to the larger increase in air temperature in northern Japan than in other regions (possibly attributed to decreasing snow cover and the prescribed ocean warming scheme in d4PDF), which allows the air to store more precipitable water in northern Japan (Ishii & Mori, 2020; Nosaka et al., 2020).

Temporally, more increase is found in return rainfall with shorter durations than the one with longer durations, ranging from 9.4% for 1-hr to 6.2% for 24-hr, averaged over the entire country of Japan and future periods (i.e., 2081–2109) (Figure 7 and Figure S6 in Supporting Information S1). This greater increase in heavy rainfall at shorter durations was also found globally in other regions like Australia, Cambodia, Netherlands, and the continental United States (Lenderink & van Meijgaard, 2008; Li, Gao, Chen, Gourley, & Hong, 2022; Prein et al., 2017; Wasko et al., 2015; Zhao, Abhishek, Kinouchi, Ang, & Zhuang, 2022, among others). This increase can be explained by the feedback in the cloud-core updrafts (Fowler et al., 2021) as well as the estimated more frequent (rapidly developing) thunderstorms (Mori et al., 2021). Notably, based on 1.2 K warming in Japan (Nosaka et al., 2020), our study indicated an apparent CC-scaling relation ($7.8\%/K = 9.4\% \div 1.2\text{ K}$) for the

return rainfall at 1-hr, slightly greater than the one at longer durations ($5.2\%/K = 6.2\% \div 1.2\text{ K}$), considering the average value over the entire country of Japan and future periods. Our findings complement previous work (Endo et al., 2017; Hatsuzuka & Sato, 2019; Kitoh & Endo, 2019) that mainly focused on multiday scales in Japan (e.g., 5-day and 10-day). Since the ways to compute CC-scaling vary in different studies (Pendergrass, 2018), care should be taken to compare our findings and others'. For some parts of Japan, a decreased return rainfall is also projected for long durations, mainly for 12- and 24-hr (Figure 7 and Figure S6 in Supporting Information S1). This decrease in heavy rainfall at daily and longer durations was also found to be consistent with Endo et al. (2017) and Kitoh and Endo (2019). This decrease may be explained by an estimated decrease in tropical cyclone frequency in the western North Pacific (Kitoh & Endo, 2019) and the associated decrease in the total precipitation (i.e., limited water availability, see Tabari, 2020) mostly in southwestern and eastern Japan (Endo et al., 2017; Kawase et al., 2019; Nosaka et al., 2020).

Apart from the spatial and temporal difference, rarer events (signified by higher return periods) are projected to intensify more (8.3% for 25-year, averaged over Japan and all rainfall durations) than less rare ones (7.3% for 2-year), in line with previous work in other TC-affected regions (Ge et al., 2019; Zhao et al., 2021; Zhao, Abhishek, Kinouchi, Ang, & Zhuang, 2022), and work focusing on heavy daily rainfall globally (Kitoh & Endo, 2019; Scoccimarro et al., 2013). Changes in atmospheric circulation may cause this greater increase in rarer events in Japan (Pendergrass, 2018), for example, the increase in very intense TC in the western North Pacific (see Yoshida et al., 2017). However, the changes in TC activities (such as frequency and intensity) informed by d4PDF have large uncertainty (Yoshida et al., 2017), and further investigation by using enhanced model data sets is needed. Moreover, although the “change factor” approach (Equation 4) can avoid the bias problems in d4PDF, this approach can be problematic and might even lead to misleading future projections because it implicitly assumes that the patterns of the precipitation and bias are stationary against future climate changes. This “change factor” approach, trained based on past data, is less capable of extrapolation in the case of previously unobserved rainfall patterns of extreme precipitation. This problem can be aggravated in case of heavy rainfalls with shorter durations where the d4PDF model cannot even reproduce well the rainfall patterns in the current period, as shown in Figure 4.

In addition to the changes mentioned above in rainfall intensity, other characteristics of heavy rainfall are equally important but have rarely been assessed so far. For instance, the changes in long-lasting (clustering) heavy rainfall events, as indicated by the extremal index of GPD in Figure 8, reveal that more clustering heavy rainfall events are likely to happen in the highlands in eastern Japan, which may increase the risks of landslides here. Some preemptive strategies like relocating people in advance or updating landslide risk maps in these highlands may be suggested. These results drawn from Figure 8 can also provide additional and useful information (besides the future rainfall intensity information) on prioritizing adaptation and mitigation actions in these blue-colored areas (Figure 8).

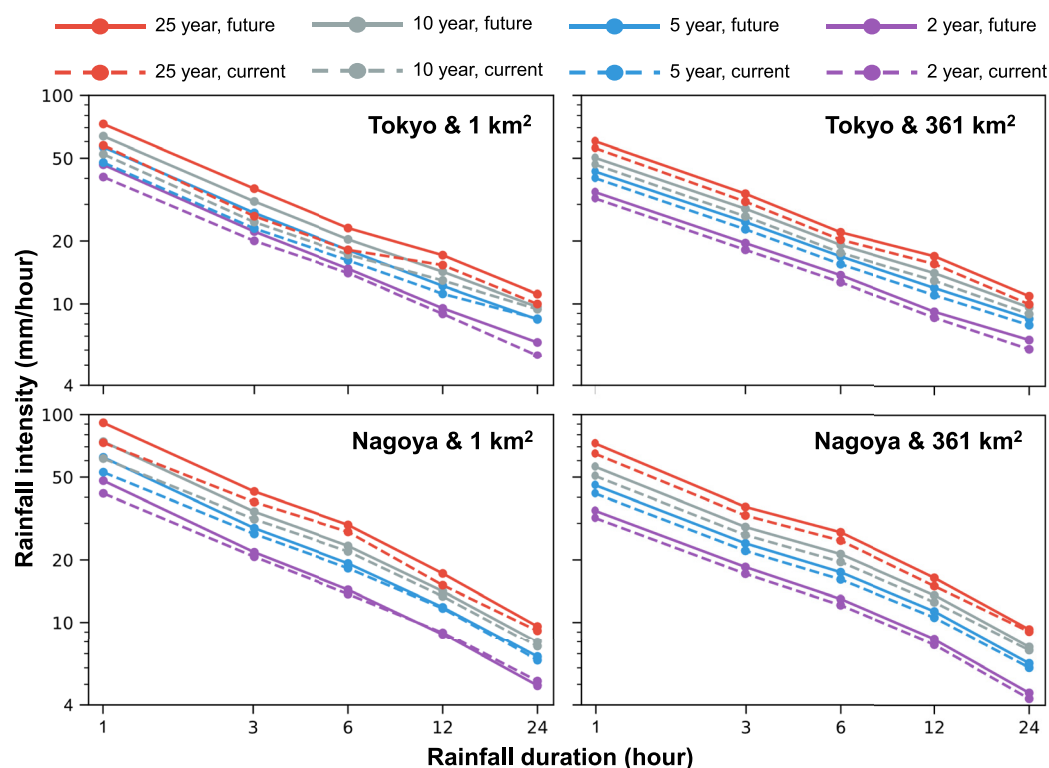


Figure 9. Current (dashed lines) and future (solid lines) intensity-duration-frequency (IDF) curves for 1 and 361 km² in (top) Tokyo and (bottom) Nagoya. The current and future IDF curves were based on the RA data and ensemble mean of 54 climate model simulations. The plots for Osaka and Kumamoto are displayed in Figure S7 of Supporting Information S1.

At last, besides CC-scaling, other mechanisms (such as complicated microphysical processes, the track and speed of TCs, interactions between rainfall generating systems, and orography) also play influential but intricate roles in heavy rainfall change, especially for the local scale events. Further investigation of the underlying mechanisms can help in quantifying and reducing the uncertainty in this study. However, this is out of scope in our present study and is the focus of future research.

4.4. Current and Future IDF Information for Different Spatial Scales

Figure 9 and Figure S7 in Supporting Information S1 display the IDF curves for 1 and 361 km² in the current (based on RA data) and future period (based on ensemble mean of d4PDF data), respectively, through the approach described in Figure 2. The associated rainfall intensities are listed in Tables S1–S4 of Supporting Information S1. Notably, the use of ARF, calculated based on past rainfall information, to obtain future IDF curves for 1 km² may be affected by the consistent pattern assumption (spatial rainfall patterns remain consistent from current to the future period). In other words, the established future IDF curves are conditioned on this assumption and may change if there are drastic changes in rainfall regimes/spatial patterns in a future climate. Although the consistent pattern assumption has been widely accepted in global warming impact studies (e.g., Hosseinzadehtalaei et al., 2021; Kay et al., 2021; Lombardo et al., 2017, among others), it needs further investigation. An alternative approach is to develop high-resolution climate models that can provide us with rainfall information at high spatial resolutions. For instance, using convection-permitting models at a resolution of several kilometers and climatic time scales cannot only provide more reliable dynamic changes (instead of the “static changes” based on ARF values) but also reduce the notable bias depicted in Figure 4 (mainly from parameterized convection) in extreme rainfalls (Li, Gao, Chen, Gourley, & Hong, 2022; Li, Gao, Chen, Gourley, Liu, 2022). We also found that, compared with IDF curves (361 km² in Figure 9) for Tokyo and Nagoya, the curves (361 km² in Figure S7 of Supporting Information S1) for Osaka and Kumamoto shift upward sharply from 3 to 6 hr. This implies that rainfall events with relatively long durations (6 hr) are more intensive in Osaka and Kumamoto, which may be caused by the more frequent mesoscale convective systems, as mentioned in Section 4.1.

Return rainfall embedded in all IDF curves (Figure 9 and Figure S7 in Supporting Information S1) generally becomes greater in the future period (against the current period), albeit with some exceptions (e.g., IDF curves for 1 km² in Nagoya and Osaka, partly owing to the decrease shown in Figure 7 and consistent pattern assumption in ARF). Specifically, return rainfall is found to increase by 8.63% in the future period, averaged for all conditions (i.e., return periods, rainfall durations, study cities, and area sizes), where rainfall intensity at a shorter duration and higher return period increase to a more extent (13.7% for 1-hr and 25-year) than the one at a longer duration and lower return period (4.8% for 24-hr and 2-year). These results generally match well with the changes displayed in Figure 7 and Figure S6 in Supporting Information S1. Moreover, return rainfall for 1 km² is projected to increase more (10.2%) than the one for 361 km² (7.1%). Besides the rainfall intensity change shown in the IDF curves, the frequency of extreme rainfall is also found to increase. For instance, in Figure 9 and Tokyo & 1 km², the rainfall intensities in 25-year IDF curves in the current period (shown as the dashed red line) are similar to the intensities in 5- and 10-year IDF curves in the future period (shown as solid blue and gray lines), meaning that the 25-year extreme rainfall events in the current period would likely be 5- and 10-year events in the future period. This finding suggests the governing agencies in these cities take tailored actions to meet specific needs for flood prevention. For instance, recently, in Osaka city, the drainage system has been updated to deal with 60 mm/hr for rainfall events with a 10-year return period (<https://www.city.osaka.lg.jp/contents/wdu020/kikikanrishitsu/english/map.html>). Such upgradation can further be continued to handle up to 61.7 and 72.7 mm/hr for 10-year and 25-year rainfall events (Table S3 in Supporting Information S1), respectively. Apart from the IDF curves for 1 and 361 km² in these case study cities, our results and approaches can be seen as an effective tool at our disposal to establish IDF curves for other spatial scales and other cities in Japan and beyond. These established IDF curves can be further transformed through the Chicago equation (Willems, 2013) into storm hyetographs, which subsequently are used to drive the hydrological models.

5. Conclusions

By using d4PDF and RA, potentially the best rainfall data (in terms of spatial (1-km and 20-km) and temporal resolutions (1-hr), number of rain-gauges (~10,000) used in RA and number of ensemble numbers (50 and 54) used in d4PDF) covering Japan, this study is the first to analyze the current and future (1.5 K warming scenario) 1–24 hr and city-scale heavy rainfall for the entire country of Japan. The POT approach was used to obtain the ARF values, which were later used for spatial downscaling of d4PDF outputs. Thereafter we quantified the climate change impacts on heavy rainfall, including the intensity, frequency, and clustering of heavy rainfall events. Major findings are summarized below:

1. We found that ARF generally decreases with the area size and increases with the duration, which is decided by relationships between the area of rainfall accumulation and the scale of storm traces. Our results further reveal that ARF varies with the return period for the rainfall durations of 1–3 hr, which can be ascribed to the impacts of convective activities on these short-duration heavy rainfall events. Moreover, elevation has a limited impact (<10%) on the variance in ARF values, with several other underlying physical phenomena (e.g., watershed location) pending further investigations.
2. Unlike the previous studies focusing mainly on daily and multi-daily scales, this study on the sub-daily scales reveals a greater increase in heavy rainfall corresponding to shorter durations and higher-return periods, ranging from 9.4% (for 1-hr) to 6.2% (for 24-hr), and 8.3% for (25-year) to 7.3% (for 2-year). Moreover, this study indicates an apparent CC-scaling relation for the rainfall at 1-hr, slightly greater than the one for rainfall at longer durations (7.8%/K for 1-hr > 5.2%/K for 24-hr). These results underscore the importance of projections for rainfall events with sub-daily scales and higher return periods.
3. Although northern Japan is characterized by a relatively small return rainfall (Figure 3), the greater increase (in these rainfalls) in this region (8.9% for northern Japan > 6.8% for other Japan areas in Figure 7) highlights the need to focus more on the hydraulic infrastructure design in northern Japan. The changes in the extremal index (Figure 8) can further help to understand whether heavy rainfall events cluster in the future or not. The results can be utilized for identifying and employing specific adaptation and mitigation actions for different locations in Japan.
4. Rainfall intensity from IDAF curves displays a greater increase in return rainfall for 1 km² (10.2%) than the one for 361 km² (7.1%). Moreover, the frequency of the extreme rainfall is also found to increase; in Tokyo & 1 km² (Figure 9), 25-year extreme rainfall events in the current period would likely be 5- and 10-year events

in the future period, which suggests the governing agencies to take tailored actions to meet specific needs for flood prevention.

Most Japanese regions are projected to be under increasing heavy rainfall stress at 1.5 K warming, as revealed by the more intense and frequent heavy rainfall. When combined with the hydrological models, the results and approaches can further help quantify the heavy rainfall-induced flood risks in different spatial scales. The dedicated assessment of the four selected cities with unique hydrogeological features and divergent socioeconomic importance corroborates the need for and the implications of our results in on-ground infrastructure developments. Regarding future work, the enhanced model data sets, which can fully resolve the TCs and convective systems, will help us better quantify the risks emerging from the changes in (mostly more intense) heavy rainfall in Japan and other east Asia regions. Finally, the climate models under CMIP6, which consider socio-economic factors in different countries, will further put forward specific investment proposals in Japan and beyond.

Data Availability Statement

The d4PDF data is publicly available from the sources cited in the main text or supplementary file. The RA data was purchased from Japan Meteorological Agency (JMA) through the Japan Meteorological Business Support Center (<http://www.jmbasc.or.jp/jp/offline/cd0100.html>, in Japanese). The data for IDF curves is listed in Tables S1–S4 of Supporting Information S1 and rainfall maps (Figures 3 and 7; Figures S1 and S6 in Supporting Information S1) are available online (<https://doi.org/10.5281/zenodo.7430620>).

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